Predicting Black Hake Catch Using Machine Learning

This project presents a machine learning model to predict the monthly catch of Black Hake in Tierra del Fuego. Inaccurate forecasting of fish stocks can cause overexploitation or underuse, harming ecosystems and local economies. Therefore, better prediction is vital for sustainable fishing. The dataset combines 2019 fishing records with local environmental variables such as sea surface temperature anomalies, precipitation, humidity, and wind. After cleaning and analyzing the data, a Random Forest Regressor was trained to make predictions. The model achieved an R-Squared score of 0.74 and a Mean Absolute Error of 750 kilograms, proving good predictive performance. These results show that climate and oceanographic factors, which may vary each year, can influence fish availability. Although the model was built with only one year of data, it demonstrates how AI can support smarter resource management. Fisheries management agencies could adopt similar models to set seasonal quotas and develop early-warning systems. Overall, this project highlights the value of machine learning to balance conservation goals and economic needs.